

Section 6.2 Subspaces and Spanning Sets

Exercise 6.2.1

Which of the following are subspaces of P_3 ? Support your answer.

a. $U = \{f(x) \mid f(x) \in P_3, f(2) = 1\}$

b. $U = \{xg(x) \mid g(x) \in P_2\}$

- e. $U =$ The set of all polynomials in P_3 with constant term 0

Exercise 6.2.3

Which of the following are subspaces of $F[0, 1]$? Support your answer.

- c. $U = \{f \mid f(0) = f(1)\}$

d. $U = \{f \mid f(x) \geq 0 \text{ for all } x \text{ in } [0, 1]\}$

g. $U = \left\{f \mid f \text{ is integrable and } \int_0^1 f(x)dx = 0\right\}$

Exercise 6.2.7

Determine whether v lies in $\text{span}\{u, w\}$ in each case.

b. $v = x; u = x^2 + 1, w = x + 2$

c. $v = \begin{bmatrix} 1 & 3 \\ -1 & 1 \end{bmatrix}; u = \begin{bmatrix} 1 & -1 \\ 2 & 1 \end{bmatrix}, w = \begin{bmatrix} 2 & 1 \\ 1 & 0 \end{bmatrix}$

Exercise 6.2.11

Let u , v , and w denote vectors in a vector space V . Show that:

b. $\text{span}\{u, v, w\} = \text{span}\{u - v, u + w, w\}$

Exercise 6.2.12

Show that $\text{span}\{v_1, v_2, \dots, v_n, 0\} = \text{span}\{v_1, v_2, \dots, v_n\}$ holds for any set of vectors $\{v_1, v_2, \dots, v_n\}$.

Section 6.3 Linear Independence and Dimension

Exercise 6.3.1

Show that each of the following sets of vectors is independent.

b. $\{x^2, x + 1, 1 - x - x^2\}$ in P_2

d. $\left\{ \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \right\}$ in M_{22}

Exercise 6.3.5

Show that the following are bases of the space V indicated.

c. $\left\{ \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \right\}; V = M_{22}$

d. $\{1 + x, x + x^2, x^2 + x^3, x^3\}; V = P_3$

Exercise 6.3.6

Exhibit a basis and calculate the dimension of each of the following subspaces of P_2 .

c. $\{p(x) \mid p(1) = 0\}$

Exercise 6.3.7

Exhibit a basis and calculate the dimension of each of the following subspaces of M_{22} .

c. $\left\{ A \mid A \begin{bmatrix} 1 & 0 \\ -1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \right\}$

Exercise 6.3.12

In each case, either prove the assertion or give an example showing that it is false.

a. Every set of four nonzero polynomials in P_3 is a basis.

h. If $\{u, v\}$ is independent, so is $\{u, u + v\}$.

i. If $\{u, v\}$ is independent, so is $\{u, v, u + v\}$.

j. If $\{u, v, w\}$ is independent, so is $\{u, v\}$.

n. If $\dim V = n$, then no set of more than n vectors can be independent.